

Digital Temperature Controller



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Digital temperature controller is an essential instrument in the field of electronics, instrumentation and control automation for measuring and controlling temperatures. It can be used as much at homes as in industrial applications. Different types of analogue and digital temperature



Fig. 1: LCD display for the temperature controller

controllers are readily available in the market, but they are generally not only expensive, their temperature range is also usually not very high. Presented here is a low-cost microcontroller-based temperature controller that can read and control temperatures in the range of zero to 1000°C. Real-time temperature is displayed on its LCD screen, and you can use it to control the temperature within the preset minimum and maximum range.

Circuit and working

Fig. 2 shows circuit diagram of the digital temperature controller. The circuit is built around microcontroller PIC-16F877A (IC1), precision thermocouple amplifier AD8495 (IC2), K-type thermocouple

(connected at CON3), 16×2 LCD (LCD1), single-changeover relay (RL1) and a few common components.

Sensor selection. There are basically two types of temperature-measurement systems—direct temperature-measurement systems for up to 1000°C and indirect temperature-measurement systems for higher temperature range, where temperature sensors may get physically damaged due to the high temperatures. The selection of temperature sensor is dependent on the range of temperature you wish to check.

TABLE I
Different Types of Direct Measurement Sensors

	Thermocouple	RTD	Thermistor
Range (°C)	-200 to 2000	-250 to 850	-100 to 300
Accuracy (°C)	>1	0.03	0.1
Thermal response	Fast	Slow	Medium
Cost	Low	High	Low to moderate
Long-term stability	Low	High	Medium

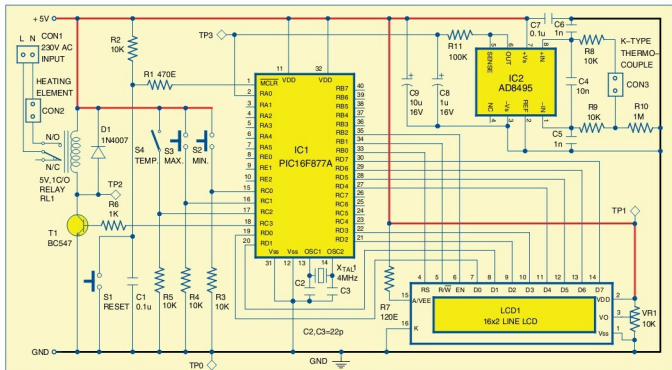


Fig. 2: Circuit diagram of the digital temperature controller